

3. Check conditions. Time windows, presence, “after sunset,” etc., can block runs silently.
4. Check delays/overlaps. A long delay can make it look “stuck.”
5. Simplify once. Temporarily remove extra logic until it triggers reliably, then add logic back.

Home Assistant superpower: Traces

Home Assistant’s official automation troubleshooting guide recommends using Traces to see step-by-step what happened after an automation ran, and it describes testing full action sequences and testing conditions/actions separately.

HA step-by-step

1. Open the automation → look for Traces.
2. Find the last run and see exactly which step failed (condition false, action error, etc.).
3. Use “Run actions” to confirm actions work even when the trigger fails.

This is the fastest way to turn “it didn’t work” into “condition blocked it at 11:32 PM because presence was ‘away.’”

Protocol drills: fix the common offenders fast

A) Wi-Fi device goes offline (one device)

1. Power-cycle the device.
2. Ensure it’s on the correct SSID (especially if you use an IoT lane).
3. Move it temporarily closer to the router to rule out coverage.
4. If it only supports 2.4 GHz, ensure your onboarding wasn’t done while your phone was on 5 GHz (classic pairing fail).



B) Zigbee sensors “drop” or report unavailable

Zigbee has one key rule: mains-powered devices strengthen the mesh; battery devices mostly “sleep.”

If many Zigbee devices go offline together

1. Check coordinator power/USB stability.
2. Restart your Zigbee stack (ZHA or Zigbee2MQTT) once.
3. Check for interference (USB 3.0 noise, router too close to coordinator).

YOUR TROUBLESHOOTING CHEAT SHEET

■ If one device is offline:

Power → SSID/band → range → firmware → reset (last)

■ If many devices are offline:

Modem/router → hub/controller → protocol stack (Zigbee/Thread) → only then re-pair

■ If automations fail but manual control works:

Trigger proof → condition audit → run actions → platform logs/traces

Zigbee2MQTT: enable availability properly

Zigbee2MQTT has an official Device Availability feature that publishes online/offline state; you enable it in configuration.

This helps you detect real dropouts vs “it’s sleeping.”

If one battery Zigbee device is “unavailable”

- Wake it up (button press) and wait; sleepy devices don’t chat constantly.
- If it still won’t rejoin, re-pair it.

If Zigbee2MQTT crashes or won’t start

Zigbee2MQTT’s official troubleshooting notes some older adapters can crash and may need reflashing or upgrading to a more powerful adapter.

C) Matter-over-Thread devices go offline

Thread devices live on a mesh, but they still depend on a Thread border router to connect to your IP network. OpenThread’s guide is clear: a Thread Border Router connects a Thread network to Wi-Fi/Ethernet, and a Thread network requires it to reach other networks.

Step-by-step

1. Confirm your border router is online (Apple TV / HomePod mini / Nest hub / HA OTBR).
2. Power-cycle the border router once.
3. If you recently added a second border router, you may have “split” Thread networks – keep one primary while things stabilize. (Thread 1.4 aims to improve shared networks, but real-world rollout varies.)

The “three reboots” rule (and why it works)

When something is truly stuck, do these in order:

1. reboot router
2. reboot hub/controller
3. reboot the device

Apple’s official Home troubleshooting essentially follows this hierarchy (network gear, hubs/bridges, then accessories). It works because you’re restarting dependencies from the foundation upward. 